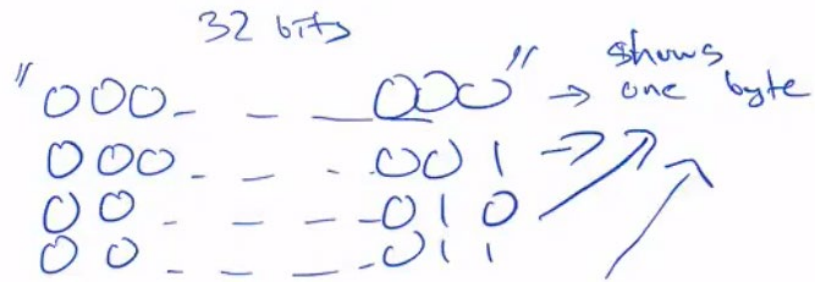
32-bit logical address space and 34 bits physical address space

a) What is the size of LAS in bytes?
2^{32} bytes = 4GB



b) What is the maximum physical memory?
PAS 34 bits => 2^{34} bytes = 16GB

c) Suppose that page is 4KB. How many pages do we have?

$$\frac{LAS\ Size}{Page\ Size} = \frac{2^{32}}{2^{12}} = 2^{20}$$

d) What is the number of entries in page table?
2^{20}

e) What is the size of an entry in page table?
34 bits but round it to nearest byte so 5 bytes

f) Size of the page table

$$2^{20}\ entries\ \times 5\ bytes$$